

Chapter 4-5 Homework B (Friction & Forces at an Angle)



System	μ_s	μ_k
Rubber on dry concrete	1.0	0.7
Rubber on wet concrete	0.7	0.5
Wood on wood	0.5	0.3
Waxed wood on wet snow	0.14	0.1
Metal on wood	0.5	0.3
Steel on steel (dry)	0.6	0.3
Steel on steel (oiled)	0.05	0.03
Teflon on steel	0.04	0.04
Shoes on wood	0.9	0.7
Shoes on ice	0.1	0.05
Ice on ice	0.1	0.03
Steel on ice	0.04	0.02

Table 1: (OpenStax Table 5.1) Coefficients of Static and Kinetic Friction

Equations

$$\Sigma \vec{F} = m\vec{a}$$

$$F_G = mg$$

$$F_{f,s} \leq \mu_s F_N$$

$$F_{f,k} = \mu_k F_N$$

Selected Answers

1. 5.00 N

2. (a) 588 N; (b) 1.96 m/s²

3. (a) -2.94 m/s²; (b) 3.29 m

4. (b) 0.70 m/s²; (c) 91 N

5. (a) 0.302 m/s², 435 N; (b) 0.302 m/s², 447 N

6. 0.737 m/s²

7. 3.67 m/s²

10. (a) $\mu_s = 0.60$ (b) $\mu_k = 0.53$

11. 104 N; $\mu_k = 0.475$

12. 5.71°

Questions and figures are from OpenStax College Physics, 2nd ed (Chapter 5) unless otherwise noted.

Friction

1. (P1) A physics major is cooking breakfast when he notices that the frictional force between his steel spatula and his Teflon frying pan is only 0.200 N. Knowing the coefficient of kinetic friction between the two materials, he quickly calculates the normal force. What is it? (Use the coefficients of friction provided in Table 1 on the front of this packet)
2. (P4) Suppose you have a 120-kg wooden crate resting on a wood floor. (a) What maximum force can you exert horizontally on the crate without moving it? (b) If you continue to exert this force once the crate starts to slip, what will the magnitude of its acceleration then be? (Use the coefficients of friction provided in Table 1)
3. (Original Question) A 7-kg wood box is pushed across a horizontal wood floor. (a) Once there is no longer any applied force, what is the box's acceleration? (b) How far does the box slide before coming to a complete stop if its initial velocity is 4.4 m/s?
4. (Original question) A metal block of mass $m_1 = 25$ kg rests on a horizontal wood table. The block is connected by a light, inextensible string that passes over a frictionless pulley to a hanging block of mass $m_2 = 10$ kg.
 - (a) Draw a free-body diagram for each block.
 - (b) Determine the acceleration of the system.
 - (c) Find the tension in the string.

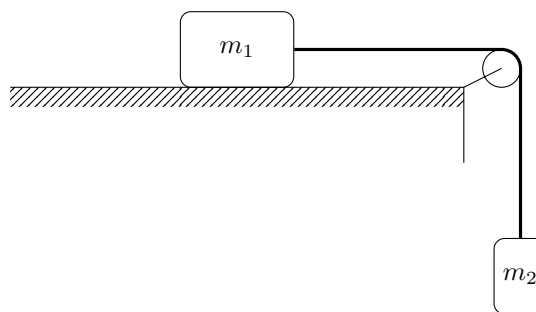


Figure 1: (Original Diagram) Figure for problem 4.

Forces at an Angle

5. (P21 with significant modification) A contestant in a winter sporting event pushes a 45.0-kg block of ice across a frozen lake as shown in Figure 2(a). A second contestant pulls it as in Figure 2(b). Both contestants exert a force $F = 15$ N. Calculate the acceleration of and normal force on each block.

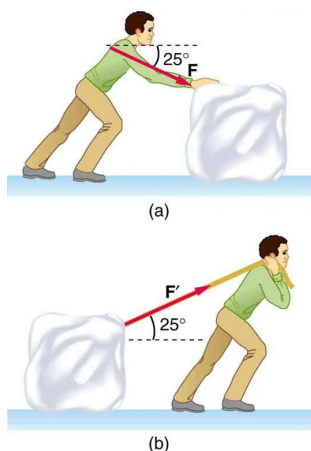


Figure 2: (OpenStax Fig 5.21) Which method of sliding a block of ice requires less force—(a) pushing or (b) pulling at the same angle above the horizontal?

6. (P11a) Calculate the acceleration of a 70-kg skier heading down a 10° slope, assuming the coefficient of friction for waxed wood on wet snow from Table 1.
7. (Ch 4, P57 from Giancoli *Physics: Principles with Applications*, 7th ed.) The block shown in Figure 3 has mass $m = 7.0$ kg and lies on a fixed smooth frictionless plane tilted at an angle $\theta = 22^\circ$ to the horizontal. Determine the acceleration of the block as it slides down the plane.

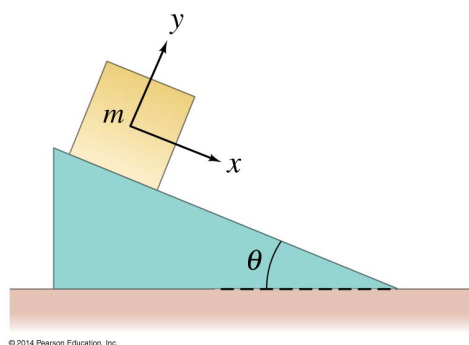


Figure 3: Fig 4-59 from Giancoli

Conceptual Questions

8. (Original Question) Look above at Figure 2 on the previous page. This time, assume that friction is *not* negligible. Which method of moving the block would be easier: pulling from in front or pushing from the back? Explain.
9. (Original Question) One block sits at rest on a level surface while a second identical block sits at rest on an incline. In which case is the normal force larger? Why?

Additional Practice

These problems are not required and are not bonus. Worked out solutions are available on Schoology.

10. (Ch 4, P37 from Giancoli *Physics: Principles with Applications*, 7th ed.) A force of 35.0 N is required to start a 6.0-kg box moving across a horizontal concrete floor. (a) What is the coefficient of static friction between the box and the floor? (b) If the 35.0-N force continues, the box accelerates at 0.60 m/s^2 . What is the coefficient of kinetic friction?
11. (Ch 4, P56 from Giancoli) A 25.0-kg box is released on a 27° incline and accelerates down the incline at 0.30 m/s^2 . Find the friction force impeding its motion. What is the coefficient of kinetic friction?

Bonus

You will be given bonus if you complete these problems *on a separate sheet of paper*

12. (P11b) A 70-kg skier is heading down a slope. Assuming the coefficient of friction for waxed wood on wet snow from Table 1, find the angle of the slope down which the skier could coast at a constant velocity.
13. (P12) If an object is to rest on an incline without slipping, friction must equal the component of the weight parallel to the incline. Show that the maximum angle of an incline above the horizontal for which an object will not slide down is $\theta = \tan^{-1}(\mu_s)$. Assume $a = 0$ and static friction has reached its maximum value.